

1. Administrative & Study Identification

Full Study Title: A Study Evaluating Single-agent Inavolisib and Inavolisib Plus Atezolizumab in PIK3CA-Mutated Cancers **Official Title:** A Phase I/Ib Study Evaluating Single-Agent Inavolisib and Inavolisib Plus Atezolizumab in PIK3CA-Mutated Cancers **Short Title/Acronym:** C043909 **Protocol Number:** C043909 (EudraCT: 2021-006618-36) **NCT Number:** NCT06496568 **Sponsor/Company Name:** F. Hoffmann-La Roche Ltd / Roche **Investigational Product:** Inavolisib (PI3K inhibitor), Atezolizumab (Tecentriq) **Phase of Development:** PHASE1 (Phase I/Ib) **Trial Status:** RECRUITING **Medical Monitor:** Roche Clinical Operations Lead [Contact Information on File]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: The purpose of the study is to assess the safety and efficacy of inavolisib as a single-agent and in combination with atezolizumab in participants with phosphatidylinositol 4,5-bisphosphate 3-kinase catalytic subunit alpha isoform (PIK3CA)-mutated cancers, including previously treated head and neck squamous cell carcinoma (HNSCC). **Secondary Objectives:** To evaluate preliminary anti-tumor efficacy endpoints including objective response rate and duration of response.

2.2 Background & Rationale PIK3CA is a gene that controls a signaling pathway (PI3K) telling cells to grow and multiply. In cancer, cells can lose the ability to turn this message off. Better treatments are needed for PIK3CA-mutated cancers that are recurrent or metastatic, such as Head and Neck Squamous Cell Carcinoma (HNSCC). This study tests inavolisib, a medicine designed to block the PI3K cancer pathway, alone or in combination with atezolizumab, which helps the immune system recognize and fight cancer cells.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Parallel group design: Arm A [inavolisib alone], Arm B [inavolisib + atezolizumab]) **Blinding:** Open-label **Randomization:** N/A (Participants are assigned to a treatment arm in the order in which they are enrolled)

3.2 Planned Enrollment & Duration Targeted Enrollment: 40 participants **Number of Sites:** Up to 15 sites across 5 countries **Estimated Study Start:** Q3 2024 **Estimated Primary Completion:** [Estimated 2026/2027]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Histologically or cytologically confirmed recurrent and/or metastatic HNSCC that has been previously treated with systemic therapy in the recurrent and/or metastatic setting. 3. Documented positive or negative human papillomavirus (HPV) status as determined locally by p16 immunohistochemistry (IHC; preferred), in situ hybridization, and/or by polymerase chain reaction-based assay. 4. Eligible participants must not be suitable for treatment with surgery and/or radiation. 5. Confirmation of biomarker eligibility: Valid results from either central testing of blood or local testing of blood or tumour tissue documenting PIK3CA-mutated tumour status. 6. Consent to provide fresh (preferred) or archival tumour tissue specimen. 7. Negative hepatitis B surface antigen (HBsAg) and total hepatitis B core antibody (HBcAb) test or positive total HBcAb test followed by a negative hepatitis B virus (HBV) DNA at screening. 8. Negative hepatitis C virus (HCV) antibody test at screening, or positive HCV antibody test followed by a negative HCV RNA test at screening. 9. Measurable disease per RECIST v1.1. 10. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 11. Life expectancy of $\geq$ 12 weeks.

4.2 Exclusion Criteria 1. Prior treatment with any PI3K, AKT, or mTOR inhibitor, or any agent whose mechanism of action is to inhibit the PI3K/AKT/mTOR pathway. 2. Appropriate for treatment with surgery and/or radiation at the time of entry. 3. Type II diabetes requiring ongoing systemic treatment at study entry; or any history of Type I diabetes. 4. Malabsorption syndrome or other condition that would interfere with enteral absorption. 5. Known and untreated, or active central nervous system (CNS) metastases (progressing or requiring anticonvulsants or corticosteroids for symptomatic control). 6. Uncontrolled pleural effusion, pericardial effusion, or ascites requiring recurrent drainage twice per week or more frequently. 7. Serious infection requiring IV antibiotics within 7 days prior to Day 1 of Cycle 1. 8. Treatment with a live, attenuated vaccine within 4 weeks prior to initiation. 9. Any concurrent ocular or intraocular condition requiring medical/surgical intervention, or active inflammatory/infectious eye conditions (e.g., uveitis, conjunctivitis). 10. Requirement for daily supplemental oxygen or symptomatic active lung disease, including pneumonitis. 11. History of or active inflammatory bowel disease (IBD) or active bowel inflammation. 12. Known Human Immunodeficiency Virus (HIV) infection. 13. Current severe, uncontrolled systemic disease or active/uncontrolled pulmonary, metabolic, or cardiovascular dysfunction. 14. Chemotherapy, radiotherapy, or any other anti-cancer therapy within 2 weeks before enrolment, or investigational drug(s) within 4 weeks. 15. Unresolved toxicity from prior therapy, except for hot flashes, alopecia, and Grade $\leq$ 2 peripheral neuropathy. 16. History of other malignancy within 5 years prior

to screening (with specified exceptions). 17. Chronic corticosteroid therapy of ≥ 10 mg of prednisone per day (or equivalent). 18. Allergy or hypersensitivity to components of the inavolisib formulation. 19. Treatment with strong CYP3A4 inducers or inhibitors within 1 week or five drug-elimination half-lives. 20. Major surgical procedure or significant trauma within 28 days prior to Day 1 of Cycle 1; minor surgical procedures < 7 days prior to first dose.

Exclusion criteria specific to arms utilizing atezolizumab: 21. Prior serious immune-mediated toxicities resulting from treatment with any checkpoint inhibitor. 22. Treatment with any checkpoint inhibitor within 5 half-lives of Day 1 of Cycle 1. 23. Uncontrolled or symptomatic hypercalcemia. 24. Active or history of autoimmune disease or immune deficiency. 25. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, idiopathic pneumonitis, or evidence of active pneumonitis on screening CT. 26. Active tuberculosis. 27. Severe infection within 4 weeks prior to initiation, including hospitalization for infection, bacteremia, or severe pneumonia. 28. Treatment with therapeutic oral or IV antibiotics within 2 weeks prior to initiation (prophylactic antibiotics may be eligible). 29. Prior allogeneic stem cell or solid organ transplantation. 30. Current treatment with anti-viral therapy for HBV. 31. Treatment with systemic immunostimulatory agents within 4 weeks or five drug-elimination half-lives. 32. Treatment with systemic immunosuppressive medication within 2 weeks prior to initiation. 33. Poor peripheral venous access precluding repeated IV infusions. 34. History of severe allergic anaphylactic reactions to chimeric or humanized antibodies, or known hypersensitivity to Chinese hamster ovary (CHO) cell products or any component of the atezolizumab formulation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Inavolisib single-agent or combined with Atezolizumab. Administered in 21-day cycles. **Route of Administration:** Oral (inavolisib) and IV Infusion (atezolizumab). **Duration of Treatment:** Continuous in 21-day cycles until disease progression, intolerable toxicity, or patient withdrawal. Total time ranges from 1 day to more than 24 months.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease symptoms or treatment-related side effects. Prophylactic antibiotics. **Prohibited:** Systemic immunosuppressive medication within 2 weeks prior to initiation; therapeutic oral or IV antibiotics within 2 weeks; systemic immunostimulatory agents within 4 weeks; strong CYP3A4 modifiers within 1 week; live, attenuated vaccines within 4 weeks.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -28 to 0	Informed Consent, Medical History, Tumor/Genetic testing (PIK3CA status via central/local test), HPV testing, Viral Panels (HBV, HCV, HIV), Baseline Scans
Baseline	Cycle 1 Day 1	First Dose of Investigational Product(s), Safety Labs
Interim	Every 21 days	Onsite Visits: Safety Check-ups, PK sampling, Efficacy/Tumor Assessments (Scans), IV infusions
End of Study / Follow-up	Varies	Monthly visits for 3 months post-treatment, then every 2 to 3 months until start of another treatment or disease progression

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Percentage of Participants with Select Treatment-related Toxicities (TRT) in Arm B. 2. Percentage of Participants with Adverse Events (AEs) and serious adverse events (SAEs).

7.2 Secondary Outcomes 1. Objective Response Rate (ORR). 2. Best Objective Response (BOR). 3. Duration of Response (DOR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular clinical check-ups. Specific monitoring for known inavolisib toxicities (hyperglycemia, severe diarrhea, nausea, rash, mucositis) and immunotherapy toxicities (immune-related AEs, infusion-related reactions). **Serious Adverse Events (SAEs):** Reporting timelines typically 24 hours for life-threatening events or hospitalizations. **Data Monitoring Committee (DMC):** Internal safety review committees routinely assess toxicities during the Phase I dose-escalation/-expansion phases.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive ≥ 1 dose of study drug); Efficacy Evaluable Set. **Sample Size Justification:** A targeted enrollment of 40 participants is determined to be adequate for a Phase I/Ib study to estimate the safety profile, maximum tolerated dose, and preliminary anti-tumor activity. **Handling of Missing Data:** Standard Phase I methodologies apply (e.g., non-responders for binary efficacy endpoints if missing).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and centralized remote monitoring for adherence to the 21-day cycle schedule, drug accountability, and AE reporting. **Audit Trail:** Electronic Case Report Form (eCRF) locking and standard source data verification processes implemented by the Sponsor.

11. Study Identifiers & Governance

Primary ID: C043909 **Registry Number:** NCT06496568 **Sponsor/Lead Organization:** F. Hoffmann-La Roche Ltd / Roche **Study Title:** A Study Evaluating Single-agent Inavolisib and Inavolisib Plus Atezolizumab in PIK3CA-Mutated Cancers **Official Regulatory Title:** A Phase I/Ib Study Evaluating Single-Agent Inavolisib and Inavolisib Plus Atezolizumab in PIK3CA-Mutated Cancers

12. Clinical Framework (Oncology/PIK3CA Specific)

Disease Areas/Conditions: PIK3CA-Mutated Cancers **Diagnosis Threshold:** Histologically or cytologically confirmed recurrent and/or metastatic HNSCC with PIK3CA mutation and measurable disease per RECIST v1.1. **Medical Methodology:** Targeted Therapy (PI3K inhibition via inavolisib) combined with Immunotherapy (PD-L1 inhibition via atezolizumab). **Optimization Target:** Tumor response (ORR, BOR, DOR), pathway blockade, and disease control.

13. Study Procedures & Methodology

Management Pathway: 21-day cycle administration of targeted therapy and/or immunotherapy. **Education Protocol (HCP):** Protocol-specific training on investigational product preparation, administration, and immune-mediated AE management. **Education Protocol (Patient):** Informed consent process, education on strict AE reporting (especially signs of hyperglycemia, rash, or infection). **Quality Audit Mechanism:** Routine site monitoring, PK data review, and central radiographic tumor assessment.

14. Observation & Follow-Up Schedule

Total Duration: Up to \$>24\$ months, dependent on patient response and disease progression. **Onsite Visit Cadence:** Every 21 days (Cycle Day 1) during active treatment. **Remote Monitoring Frequency:** As needed to monitor toxicities or follow up on AEs. **Checklist Review Interval:** Monthly for 3 months post-treatment, followed by Q2M or Q3M long-term follow-up scans.

Role	Name	Signature	Date		:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					